

Rudder removal from an AMEL Super Maramu 2000

CONTENT

1	FIGURES	3
2	AS-IS SITUATION	5
3	PREPARATION FOR RUDDER REMOVAL	7
3.1	OUTSIDE PREPARATION	7
3.1.1	STARBOARD SIDE	7
3.1.2	PORT SIDE	9
3.2	INSIDE PREPARATION	10
4	RUDDER REMOVAL	10
4.1	OUTSIDE	10
4.2	INSIDE	16
5	THE REPAIR	18
6	PAINTING	20
7	RE-INSTALLATION	21
7.1	OUTSIDE	21
7.2	INSIDE	30
8	FINAL WORK	31

1 FIGURES

Figure 1 - Cracks in the rudder at the bottom on starboard side	5
Figure 2 - Cracks in the rudder at the bottom on port side	6
Figure 3 - Cracks in the rudder at the bottom on starboard side which could be easily opened with a screw driver.....	6
Figure 4 - Stainless steel pintle bearing (middle and bottom) of starboard side.....	7
Figure 5 - Bottom bearing with welded wire on the mounting bolt	8
Figure 6 - Stainless steel pintle bearing (middle and bottom) of port side.....	9
Figure 7 - The quadrant in the aft cabine which has to release from the top shaft	10
Figure 8 - Small bolts at the U-shaped middle bearing stainless steel stripe	11
Figure 9 - Support the rudder with wooden blocks and a vehicle jack	11
Figure 10 - The opened up U-shaped middle bearing stainless steel stripe	12
Figure 11 - Removing the rudder carefully.....	13
Figure 12 - The removed rudder. Ready for detailed inspection/examination	14
Figure 13 - Removed rudder with the bottom mounting and the wire with welded bolt.....	15
Figure 14 - Super Maramu 2000 Redline w/o rudder	15
Figure 15 - The removed quadrant and the nylon nut.....	16
Figure 16 - Counter plate which has to unscrew first to remove the nylon nut	16
Figure 17 - I borrowed this wrench from a worker at the boatyard to unscrew the nylon nut. Later I received the AMEL tool for re-installing the nut.....	17
Figure 18 - AMEL tool for the nylon nut.....	17
Figure 19 - Removed counter part, nylon nut and shaft packing.....	18
Figure 20 - The rudder cleaned down to sane material (bottom with major cracks)	19
Figure 21 - The rudder cleaned down to sane material	19
Figure 22 - The repaired rudder with first layer of epoxy primer	20
Figure 23 - The wooden block is in position for re-installation	21
Figure 24 - Bring the rudder on the wooden block	22
Figure 25 - Bottom mounting bearing at the bottom of the rudder shaft.....	23
Figure 26 - Vehicle jack between the wooden blocks	24
Figure 27 - Applied sealant at the bearing areas.....	25
Figure 28 - Liftung up the rudder in its final position.....	26

Figure 29 - U-shaped middle bearing stainless steel stripe Squeezed with wooden planks	27
Figure 30 - Installed bolts for the bearings	28
Figure 31 - The re-installed rudder	29
Figure 32 - The re-installed rudder	30
Figure 33 - The packing has to cut at an angle	30
Figure 34 – The three shaft packings has to insert into place like above	31
Figure 35 - First layer of Epoxy primer on starboard side	32
Figure 36 - First layer of Epoxy primer on port side	32
Figure 37 - One layer of fiberglas matts on starboard side	33
Figure 38 - One layer of fiberglas matts on starboard side	33

2 AS-IS SITUATION

After sanding down the old antifouling I noticed that the rudder had some cracks at the bottom mounting.

I discussed that with the AMEL forum, with an AMEL surveyor and also AMEL directly.

The answers were different: from “it is not necessary to repair”, “we sailed with those cracks 30,000NM and nothing happened” to “it has to be repaired”. I decided to repair these cracks because the rudder is one of the most important things aboard for safety.

Cracks in the rudder at the lower mounting at the starboard side which are going around to the port side:



Figure 1 - Cracks in the rudder at the bottom on starboard side

The cracks on the port side:



Figure 2 - Cracks in the rudder at the bottom on port side



Figure 3 - Cracks in the rudder at the bottom on starboard side which could be easily opened with a screw driver

3 PREPARATION FOR RUDDER REMOVAL

3.1 OUTSIDE PREPARATION

3.1.1 STARBOARD SIDE

Sand / grind down the GRP where the stainless steel pintle bearing (bottom) and intermediate bearing are bolted through. Be careful when you reach the screws not to sand them down.

To uncover the wire between the middle and lower mounting I used a ripping chisel and hammer:



Figure 4 - Stainless steel pintle bearing (middle and bottom) of starboard side

Notice that the wire is welded at the screw (of the lower mounting). Do not cut it, remove it complete (the wire with the screw):



Figure 5 - Bottom bearing with welded wire on the mounting bolt

3.1.2 PORT SIDE

Sand/grind down the surface until you see the mountings on the port side of the skeg:



Figure 6 - Stainless steel pintle bearing (middle and bottom) of port side

3.2 INSIDE PREPARATION

In the aft cabine remove the bolts and release the quadrant to release it from the shaft. It is not necessary to unscrew the grounding cable (yellow/green) linking to copper plate:



Figure 7 - The quadrant in the aft cabine which has to release from the top shaft

4 RUDDER REMOVAL

4.1 OUTSIDE

To remove and handle the rudder easily you need as minimum 2 strong people, better would be three. All positioned outside. Inside no one is required but you can positioning an additional person inside.

Be sure that you have enough space between the rudder and the ground to allow the rudder slipping down. You need as minimum ca. 60cm clearance between the rudder and the ground.

Support the rudder with wooden blocks and a vehicle jack.

Remove all bolts of the middle and the bottom bearing.

You need to open up the U-shaped middle bearing stainless steel stripe (see “Figure 10 - The opened up U-shaped middle bearing stainless steel stripe”). For that you have to unscrew the small bolts.



Figure 8 - Small bolts at the U-shaped middle bearing stainless steel stripe

Then slowly drop down the rudder blade with the vehicle jack. Be sure that the rudder drop down w/o changing the angle/cranking the rudder until it is complete removed and you see the top of the shaft.



Figure 9 - Support the rudder with wooden blocks and a vehicle jack



Figure 10 - The opened up U-shaped middle bearing stainless steel stripe

Lay down the rudder on a pallet of some wooden blocks. Now it is ready to inspect, check for more cracks, bubbles in the GRP and/or faulty repairs etc.

I also used the chance to check if between the zink anodes mountings on port and starboard side and the bottom shaft is electrical flow. You should do that as-well to be sure that nothing of the inside connections brokes.

Also use the chance to inspect the skeg if there are also any cracks, faulty repairs etc.



Figure 11 - Removing the rudder carefully



Figure 12 - The removed rudder. Ready for detailed inspection/examination



Figure 13 - Removed rudder with the bottom mounting and the wire with welded bolt



Figure 14 - Super Maramu 2000 Redline w/o rudder

4.2 INSIDE

Under the quadrant is the nylon nut which has to unscrew. For that you have to unscrew the counter plate (see Figure 16 - Counter plate which has to unscrew first to remove the nylon nut). This counter plate at the nut is to stop it from turning and remove it.

Under the nylon nut you will find 3 layers of packing for sealing. I took the chance to renew the shaft packing (see chapter 7.2 "inside").



Figure 15 - The removed quadrant and the nylon nut



Figure 16 - Counter plate which has to unscrew first to remove the nylon nut



Figure 17 - I borrowed this wrench from a worker at the boatyard to unscrew the nylon nut. Later I received the AMEL tool for re-installing the nut.



Figure 18 - AMEL tool for the nylon nut



Figure 19 - Removed counter part, nylon nut and shaft packing

5 THE REPAIR

Do all necessary repairs using proper methods: clean down to sane material, rebuild using resin fiber cloth before protecting.

My rudder was repaired by a professional company.

When sanding down to sane material the repair shop found out that the rudder had some bubbles in the GRP. They must be from the initial building.

They used two layers of Biax fiberglass. One big layer for both sides. The layers were not cut in the front of the rudder: each fiberglass layer was placed around in the front of the rudder.



Figure 20 - The rudder cleaned down to sane material (bottom with major cracks)



Figure 21 - The rudder cleaned down to sane material



Figure 22 - The repaired rudder with first layer of epoxy primer

6 PAINTING

Before replacing back the rudder you should take the chance to paint as minimum all areas where you cannot reach when the rudder is reinstalled, i.e. the skeg where the front part of the rudder will cover it, the front side of rudder itself.

I painted the complete skeg (w/o the area which you have to laminate back) with 3 layers of epoxy primer and than one layer of antifouling.

I also organized a spray gun to spray epoxy primer into the rudder to protect the repaired areas inside. For that I used the holes in the top and bottom and the both holes for the zink anodes (port and starboard side).



7 RE-INSTALLATION

7.1 OUTSIDE

And now all the way back...

Bring back the wooden blocks in their position to support the rudder. You need again as minimum 2 strong guys, better would be 3 guys.



Figure 23 - The wooden block is in position for re-installation

Bring the rudder to the skeg and lay it on the wooden support. One person should hold the rudder all the time.



Figure 24 - Bring the rudder on the wooden block

Put the stainless steel mounting block back in place at the bottom of the shaft.



Figure 25 - Bottom mounting bearing at the bottom of the rudder shaft

Put the vehicle jack between the wooden blocks.



Figure 26 - Vehicle jack between the wooden blocks

Apply sealant at the areas where the stainless steel bearings has to mount. Use this sealant also for the bolts.



Figure 27 - Applied sealant at the bearing areas

Lift up the rudder with the vehicle jack in its correct position.



Figure 28 - Liftung up the rudder in its final position

The opened U-shaped middle bearing stainless steel stripe has to squeeze so that it is possible to mount with the bolts. We used long wooden planks tied at each ends together.



Figure 29 - U-shaped middle bearing stainless steel stripe Squeezed with wooden planks

Install the bolts back with marine sealant.



Figure 30 - Installed bolts for the bearings

Finally the rudder is re-installed.



Figure 31 - The re-installed rudder



Figure 32 - The re-installed rudder

7.2 INSIDE

Also inside your have to re-install the sealant packing, the nylon nut and the quadrant.

For the shaft packing I used 70cm of PTFE Flax shaft packing 5/16" which have to cut into three pieces. Cut the corners at an angle for a better sealing.

I never remember the length of each part but you can use the old packings or the hole for measurement.

When installing the new packing coat them with silicon grease.



Figure 33 - The packing has to cut at an angle

When taking into place be sure that the cuts are as far away from each other for better sealing:



Figure 34 – The three shaft packings has to insert into place like above

Screwing the nylon nut after each navigation until well tightened (but not too much!).

Check bolts on square section regularly, they could come unfastened with rudder motion if not checked.

8 FINAL WORK

The final work is to bring the open areas at the bearings and between in shape:

1. First I used epoxy primer to fill the area of the bearings and the wire. Sand it down to nearly the final shape.
2. Then I overlayed the areas with a fiberglass mat and sanded as-well.
3. Use the epoxy primer for a last final adjustment of the final shape.
4. Paint it with epoxy primer and than with antifouling.



Figure 35 - First layer of Epoxy primer on starboard side



Figure 36 - First layer of Epoxy primer on port side



Figure 37 - One layer of fiberglass matts on starboard side



Figure 38 - One layer of fiberglass matts on starboard side